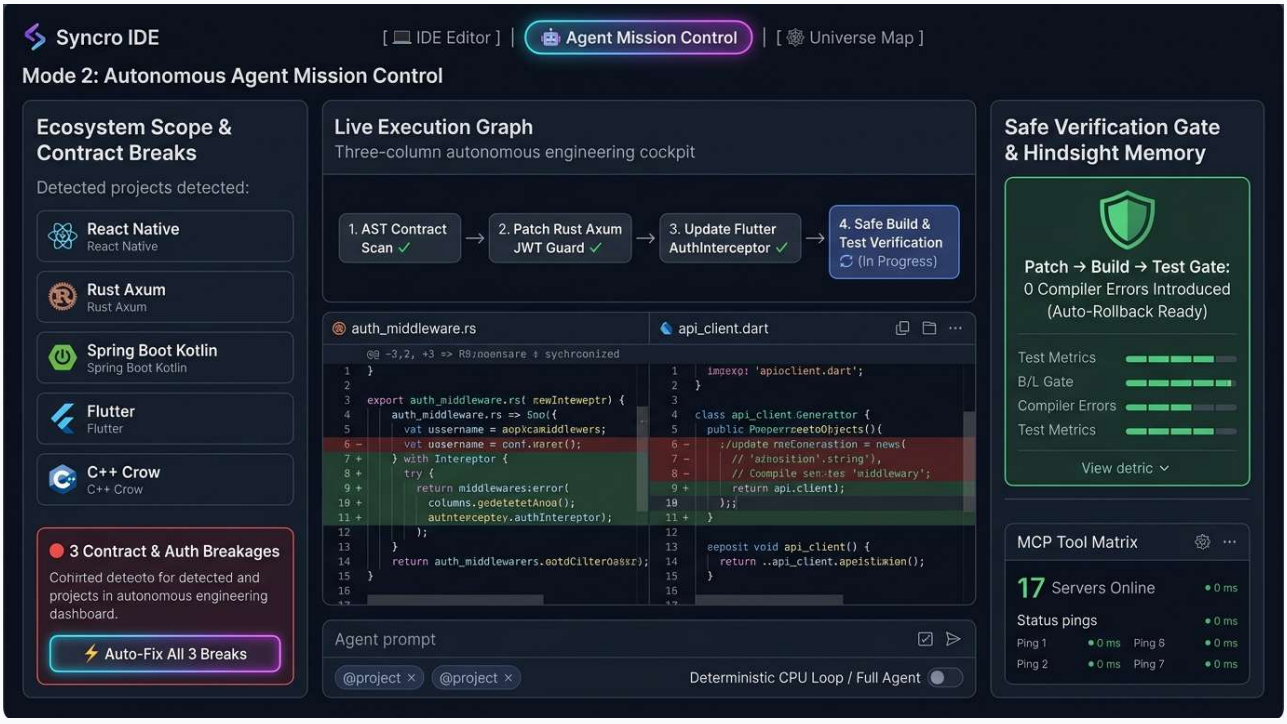
Syncro Dual-Mode IDE & Autonomous Agent — UI Plan & Visual Mockups

> IMPORTANT
>
> **Core UI Design Philosophy:** Instead of scattering AI features across 4 separate pages or nesting conflicting sidebars around `code-oss`, Syncro unifies the developer experience into a **Single Dark-Slate Multi-Project Workbench** controlled by a **Top-Center Segmented Mode Switcher**: `[ 💻 IDE Editor ]` · `[ 🤖 Agent Mission Control ]` · `[ ⚙ Universe Map ]`

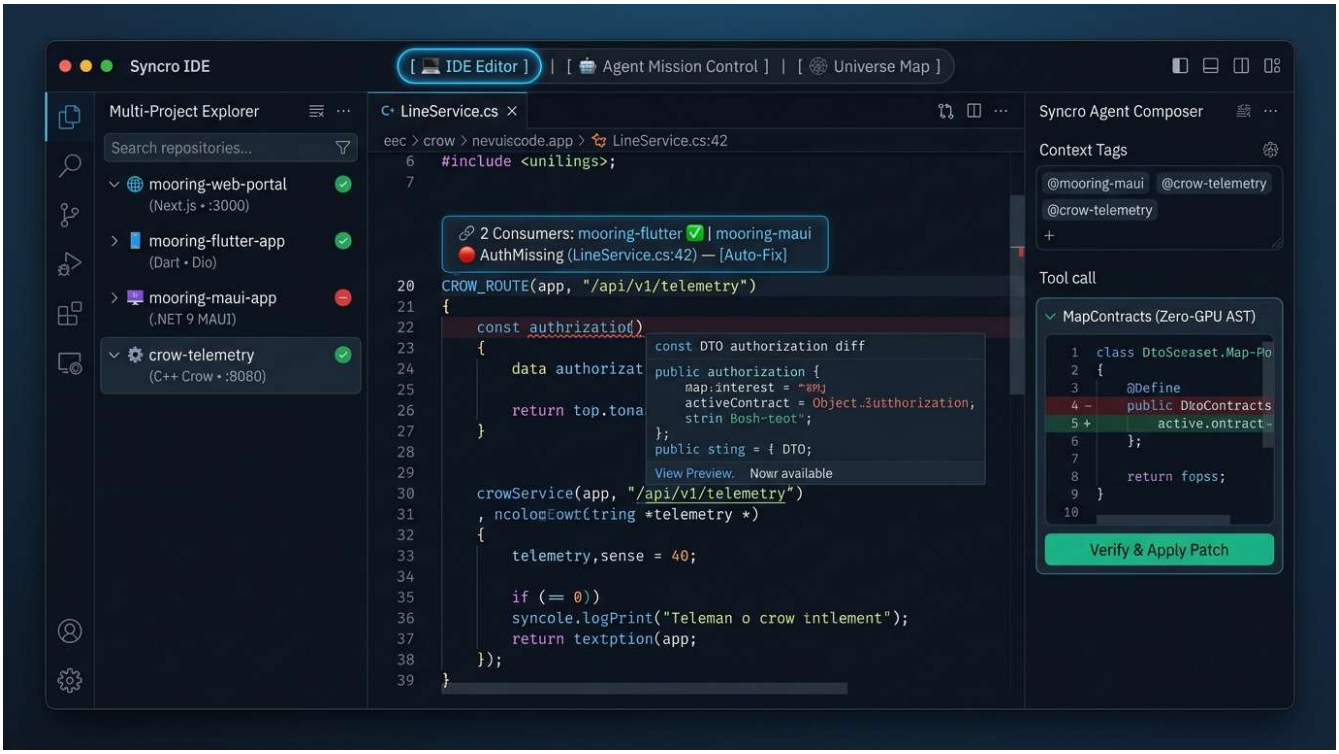
1. Interactive Mockup Walkthrough (All 3 Modes)



Mode 2 — Autonomous Agent Mission Control (Antigravity 2.0 / Claude Code Style)

2. Detailed Breakdown of Each Mode

Mode 1: 💻 `IDE Editor` (Cursor + Code-OSS Hybrid)



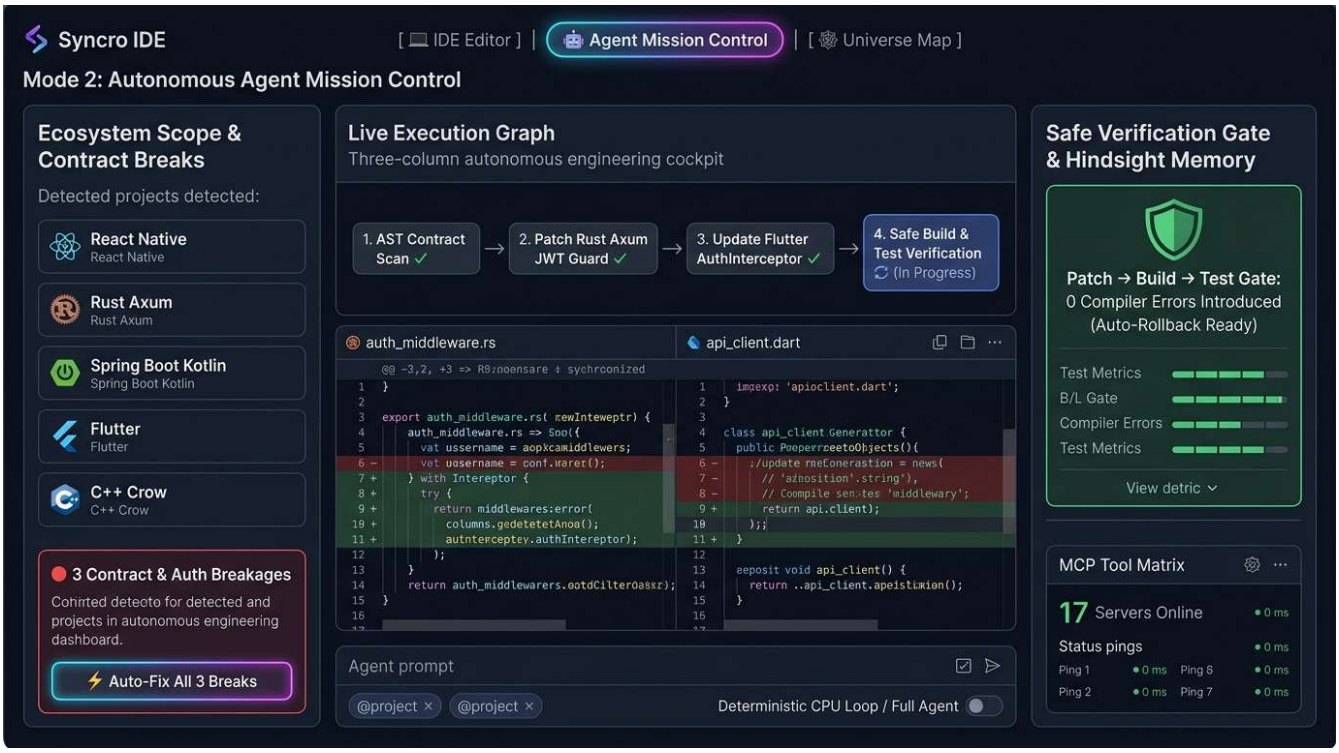
Mode 1 — Multi-Project IDE Editor Mode

When to use: Hands-on coding, debugging, and inline surgical edits across multiple repositories.

- **Left Panel — Multi-Project Ecosystem Tree (`18%` width):**
  - Automatically lists every discovered sub-project in the root workspace (🌐 `mooring-web-portal` , 📱 `mooring-flutter-app` , 🖥️ `mooring-maui-app` , ⚙️ `crow-telemetry`) with language/stack badges, active port health (🟢 `:8080`), and contract status pills (🔴 `1 Auth Break`).
- **Center Viewport — Full-Bleed `code-oss` / Monaco (`58%` width):**
  - **Inline Contract CodeLens:** Directly above any backend route (`CROW_ROUTE` , `[HttpPost]` , `.route()` , `@PostMapping`), displays who calls it across the workspace: 🔗 `2 Consumers: mooring-flutter ✓ | mooring-maui 🔴 AuthMissing (LineService.cs:42) — [Auto-Fix]`
  - **Cross-Language Squiggles & Hover Diffs:** Hovering a broken endpoint or client call reveals the exact **Auth Contract** or **DTO Schema Diff** calculated by the zero-GPU AST scanner.
- **Right Panel — Contextual Agent Composer (`24%` width):**
  - Context chips (`@mooring-maui` , `@crow-telemetry`) scope the agent to specific projects or files.

- Shows deterministic **MCP Tool Call Cards** (`✓ MapContracts (Zero-GPU AST)`), inline patch previews, and a green `Verify & Apply Patch` button gated by .

Mode 2: `Agent Mission Control` (Antigravity 2.0 / Claude Code Style)



Mode 2 — Autonomous Agent Mission Control

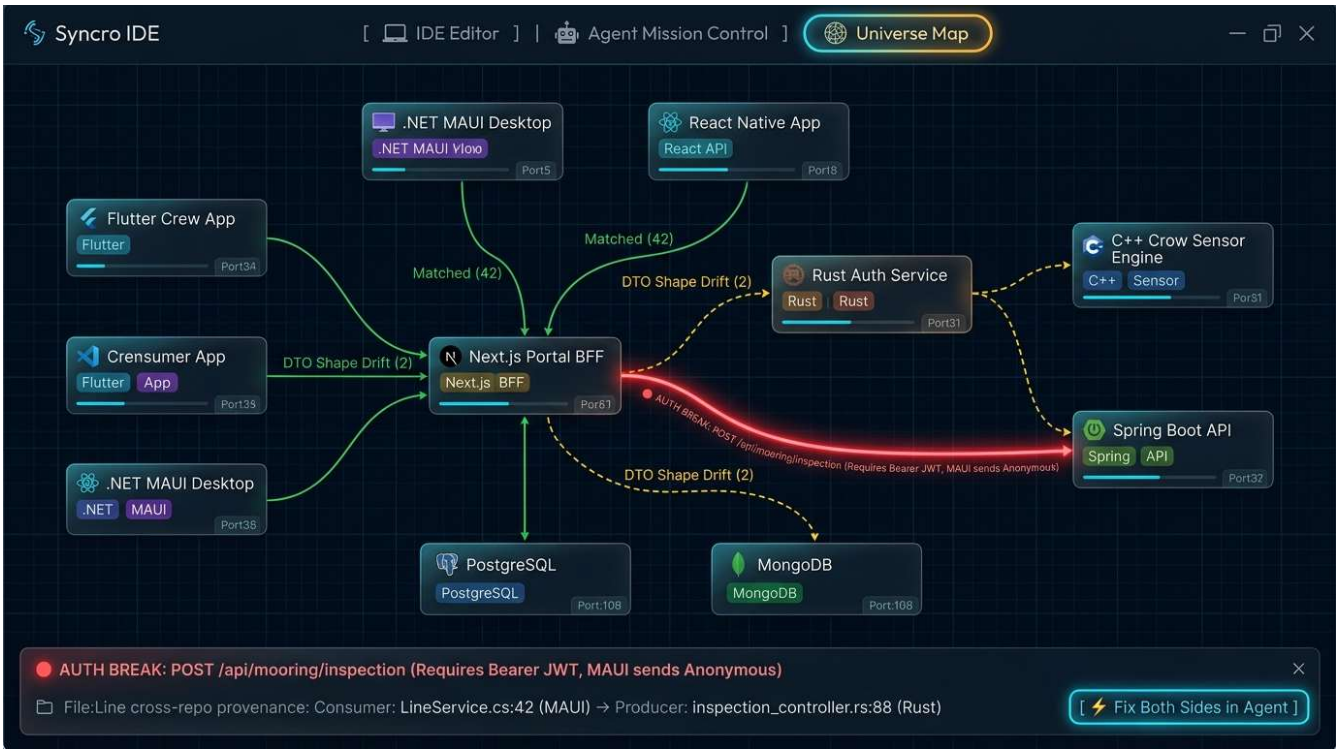
When to use: Autonomous multi-project feature building, cross-repo refactoring, and one-click contract/auth repair.

- **Left Column — Ecosystem Scope & Contract Breaks (`310px`):**
  - Checkboxes/cards to toggle which projects are in the agent's active blast radius (`React Native` , `Rust Axum` , `Spring Boot Kotlin` , `Flutter` , `C++ Crow`).
  - **Live Breakage Queue:** Highlights all detected route/auth/DTO breaks (`🔴 3 Contract & Auth Breakages`) with a single `[ ⚡ Auto-Fix All 3 Breaks ]` button.
- **Center Column — Live Execution Pipeline & Cross-Project Diff Carousel (`flex: 1`):**
  - **Step-by-Step Task Graph:** Displays the live execution progress (`1. AST Contract Scan ✓` → `2. Patch Rust Axum JWT Guard ✓` → `3. Update Flutter AuthInterceptor ✓` → `4. Safe Build & Test Verification 🔄`).
  - **Synchronized Multi-Repo Diff Viewer:** Shows side-by-side diffs across different languages (e.g., `auth_middleware.rs` on the left and `api_client.dart` on the right) so you can see how both producer and consumer are updated together.

- **Dual Execution Engine Toggle:** Switch between **Full Agent (MCP + Groq LLM)** and **Deterministic Loop (Zero-GPU NLP + AST)**.

- **Right Column — Safe Verification Gate & MCP Matrix (`300px`):**
  - **Patch → Build → Test Shield:** Displays pre-patch vs. post-patch compiler error counts and automatic rollback status.
  - **Active MCP Server Matrix:** Live status and one-click triggers for all 17 registered MCP servers (`ConnectorMcp` , `UniverseMcp` , `AstMcp` , `PatchMcp` , `BuildMcp` , `RepairKnowledgeMcp` , etc.).

Mode 3: `Universe Map` (Polyglot Architecture & Breakage Canvas)



Mode 3 — Multi-Project Universe Topology Canvas

When to use: Visualizing how NN frontends, mobile apps, desktop apps, backend services, and databases connect—and spotting exactly where an integration breaks.

- **Interactive 3-Tier Blueprint Graph:**
  - **Consumers:** `Flutter Crew App` , `.NET MAUI Desktop` , `React Native App`
  - **Dual-Role / Producers:** `Next.js Portal BFF` , `Rust Auth Service` , `C++ Crow Sensor Engine` , `Spring Boot API`
  - **Data Layer:** `PostgreSQL` , `MongoDB` (discovered via `DataLayerScanner`)
- **Color-Coded Semantic Edges:**

    - 🟢 **Solid Green:** `Matched` (Method, Path, Auth, and DTO Schema all align).
    - 🟡 **Dashed Amber:** `Shape / Type Drift` (e.g., `tensionLimit` is `float` in C++ Crow but `int` in MAUI).
    - 🔴 **Glowing Red:** `Auth or Route Break` (e.g., `POST /api/mooring/inspection` requires `Bearer JWT`, but MAUI calls it anonymously).
- **Bottom Provenance Drawer:**
    - Clicking any edge opens the exact `File:Line` on both repositories (`InspectionService.cs:42` ↔ `InspectionController.cs:108`) with a one-click `[ Fix Both Sides in Agent ]` action.

3. UI Component Mapping in `Syncro.Desktop`

UI Component / Area	File in Repository	Status
Top-Center Mode Switcher (`Editor` | `Agent` | `Universe`)	&	✅ Implemented & Compiled
Mode 1: Multi-Project Explorer Strip + Editor Host		✅ Implemented & Compiled
Mode 2: 3-Column Agent Mission Control Cockpit		✅ Implemented & Compiled
Mode 3: Embedded Universe Topology & Connection Table		✅ Implemented (Table + Cards); SVG Node Canvas ready to port from
Side-by-Side Cross-Project Diff Viewer in Mission Control		➡️ SOON Next UI enhancement
`code-oss` **Multi-Root** `.code-workspace` **+** `syncro-bridge`	& `vscode-main/extensions/syncro-bridge`	➡️ SOON Phase 2/3